

Multiple Choice (10 marks)

1. Solve the equation $14 = w + 23$.
 - (a) 9
 - (b) 37
 - (c) -23
 - (d) 14
 - (e) -14
2. When the expression $-2x^2 - 20x - 53$ is written in the form $a(x + d)^2 + e$, where a , d , and e are constants, what is the sum $a + d + e$?
 - (a) 0
 - (b) 2
 - (c) 6
 - (d) 80
 - (e) 20
3. Use mental math to solve $8y = 56$.
 - (a) 8
 - (b) 10
 - (c) 4
 - (d) 14
 - (e) 6
4. Suppose X and Y are random variables with $E(X) = 312$, $\text{var}(X) = 6$, $E(Y) = 307$, and $\text{var}(Y) = 8$. What are the expected value and variance of the random variable $X + Y$?
 - (a) $E(X + Y) = 307$, $\text{var}(X + Y) = 10$
 - (b) $E(X + Y) = 307$, $\text{var}(X + Y) = 6$
 - (c) $E(X + Y) = 619$, $\text{var}(X + Y) = 14$
 - (d) $E(X + Y) = 619$, $\text{var}(X + Y) = 7$
 - (e) $E(X + Y) = 312$, $\text{var}(X + Y) = 14$
5. Suppose today is Wednesday. What day of the week will it be $10^{(10^{(10)})}$ days from now?
 6. Sunday

7. Wednesday
8. Tuesday
9. Saturday
10. The day of the week cannot be determined

True / False (10 marks)

Answer True or False

6. True or False: Every integral domain with prime characteristic must be infinite, as seen in fields of prime order.
7. True or False: The determinant of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is 6.
8. True or False: The critical value for $F_{0.01}(8, 4)$ is 16.30, indicating the threshold for the upper 1% of the distribution.
9. True or False: The sum of all possible values of x that satisfy the equation is 3.
10. True or False: The number of ways to distribute 13 identical balls into 4 distinct boxes, with none empty, is 220.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the initial value problem $2y'' + 3y' - 2y = 0$ with conditions $y(0) = 1$ and $y'(0) = -\beta$ for $\beta > 0$. Determine the smallest value of β such that the solution lacks a minimum point, providing a detailed explanation of your reasoning and any relevant mathematical principles.
11. Consider a unit square in which 19 points are placed arbitrarily. Analyze the placement of a circle with a diameter of $\frac{\sqrt{2}}{3}$ and determine the minimum number of points that can be guaranteed to be covered. Provide a detailed explanation of your reasoning and any relevant mathematical principles.

End of Paper